Linear Equations: One, None, or Infinitely Many Solutions
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 4	5. —	9. , $4x+8$, 8
2. —	6. $17-8x$	10. —
3. $2x+6$	7. 3	
4. 7	8. $6x-3$	

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.C.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 14 tagged checks.

1. **Two saving plans:** $3w+5=2w+9$. Subtract $2w$ from both sides: $w+5=9$. Subtract 5: $w=4$. The w-terms were different ($3w$ against $2w$), so the variable survived and exactly one value works. Check: $3(4)+5=17$ and $2(4)+9=17$ dollars.

$\boxed{\text{one solution: } w=4 \text{ weeks}}$

2. **Two printers:** $4t+7=4t+2$. The table shows the totals stay 5 pages apart: at $t=1$ they are 11 and 6, at $t=2$ they are 15 and 10, at $t=3$ they are 19 and 14. Algebraically, subtracting $4t$ from both sides leaves $7=2$, which is false for every t. Identical t-terms with different constants means the two totals can never meet. $\boxed{\text{no solution}}$

3. **Perimeter two ways:** $2(x+3)=2x+6$. Distribute on the left: $2x+6=2x+6$. Both sides are the same expression, so the statement is true for every value of x — Sam and Dana wrote the same perimeter in two forms. $\boxed{2x+6=2x+6,\ \text{infinitely many solutions}}$

4. $5(x-2)=3x+4$. Distribute: $5x-10=3x+4$. Subtract $3x$: $2x-10=4$. Add 10: $2x=14$. Divide by 2: $x=7$. The x-terms differed, so there is exactly one solution. Check: $5(7-2)=25$ and $3(7)+4=25$. $\boxed{\text{one solution: } x=7}$

5. $6x-3(x+2)=3x+1$. Distribute on the left: $6x-3x-6=3x+1$. Combine like terms: $3x-6=3x+1$. Subtract $3x$: $-6=1$, which is false. The left and right sides have the same x-term but different constants, so no value of x can work. $\boxed{\text{no solution}}$

6. $-2(4x-6)+5=17-8x$. Distribute: $-8x+12+5=17-8x$. Combine the constants on the left: $-8x+17=17-8x$. The two sides are the same expression written in a different order, so the equation is true for every x. $\boxed{17-8x=17-8x,\ \text{infinitely many solutions}}$

7. **Two gyms:** $25 + 15m = 40 + 10m$. Subtract $10m$: $25 + 5m = 40$. Subtract 25: $5m = 15$. Divide by 5: $m = 3$. The monthly rates differ (15 against 10 dollars), so the costs cross exactly once. Check: $25 + 15(3) = 70$ and $40 + 10(3) = 70$ dollars.

$\boxed{\text{one solution: } m = 3 \text{ months}}$

8. $3(2x - 1) = ax + 7$ with no solution.

(a) Distributing the left side gives $\boxed{6x - 3}$

(b) No solution requires the two sides to have the *same* x-term, so ax must equal $6x$.

$\boxed{a = 6}$

(c) **Model explanation (open response — review by hand).** With $a = 6$ the equation reads $6x - 3 = 6x + 7$. Subtracting $6x$ from both sides removes the variable entirely and leaves $-3 = 7$, a statement that is false no matter what x is. Matching x-terms is what erases the variable; different constants are what make the leftover statement false. Both conditions are needed: if the constants had matched as well, every x would work instead of none.

9. Classifying by inspection.

(a) $8x - 3 = 8x + 1$: same x-term, different constants. Subtracting $8x$ leaves $-3 = 1$, false.

$\boxed{\text{no solution}}$

(b) $4(x + 2) = 4x + 8$: distributing the left gives $4x + 8$, identical to the right.

$\boxed{4x + 8 = 4x + 8, \text{ infinitely many solutions}}$

(c) $5x - 6 = 3x + 10$: different x-terms, so the variable survives. Subtract $3x$: $2x - 6 = 10$; add 6: $2x = 16$; divide by 2: $x = 8$. $\boxed{\text{one solution: } x = 8}$

10. **Two ride prices:** $2n + 3 = 2n + 5$.

(a) Subtracting $2n$ from both sides leaves $3 = 5$, which is false, so no ride length n makes the prices equal. $\boxed{\text{no solution}}$

(b) **Model explanation (open response — review by hand).** Both companies charge the same amount per mile, so the gap between their prices is the difference in the fixed charges and never changes: Company Q always costs exactly 2 dollars more, whatever the distance. There is no ride length — short, long, or fractional — at which the totals agree, which is exactly what "no solution" says. A rider should simply always choose Company P.